EL NIÑO ACTION BRIEFING · 30 OCTOBER 2026

Maroua, Extrême-Nord

Most of the Far North's forecast heat deaths are still ahead, with 910 in March alone, so rehydration salts belong in the quarters, and a named neighbour should be checking on the isolated, before the hot season peaks.

REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Extrême-Nord Region (GADM CMR.4_1), region level	2,500 forecast Oct 2026 – Mar 2027 (October 590, November 510, March 910); 490 already past (Jun–Sep)	6-month forecast, Oct 2026 – Mar 2027	Heat (Oct–Nov and Mar–May); meningitis Dec–June; cholera with the rains; power and water cuts; the 2027 lean season

22/22 recommendations checked. 3 confirmed, 19 corrected, 0 unverified, 0 contraindicated among the recommendations that remain. That covers the 9 ranked actions (all corrected), the 4 "Do not do this" cards (1 confirmed, 3 corrected), and the 9 For you steps (2 confirmed, 7 corrected). One piece of advice the panel drafted, to wet the skin and sit in front of a fan, was found harmful above about 38 °C in dry heat. It is now the first "Do not" card. Exposure figure: Climate Impact Lab El Niño Heat Mortality Tracker data, version of 26 October 2026, pre-peer-review. It is published for Extrême-Nord Region, not for the city, and counts heat only. No city share was calculated. Few deaths in the region are registered, and no heat death has been officially counted, so the forecast cannot be checked against a local count. After verification, the funding target fell from XAF

EXECUTIVE SUMMARY — AT A GLANCE



10.59°N 14.32°E, Cameroon · Natural Earth

- **The risk.** 2,500 forecast heat deaths in the region by March, plus meningitis and cholera; no heat death has ever been officially counted here.
- **For you.** By 30 November, keep rehydration salts and zinc at home for every child under five, and restock them by 15 February.*
- **The money.** XAF 97–370 million (US$165,000–630,000); cash to the poorest households in February buys the most.
- **Do not.** Above about 38 °C, do not sit in front of a fan, even with wet skin: wet the skin with the fan off.*

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* Suggested formulation: check with a health worker before sharing.

264.2 million (US$450,000) to XAF 184.9 million (US$315,000), down 30%. The Climate Impact Lab Adaptation Inventory loads in the browser and returned no entries to the research agents, so no precedent could be confirmed there.

FOR YOU, AS A RESIDENT

9 STEPS

For a family in Maroua, this El Niño shows up as two hot spells, one now and a worse one in March, with power cuts during them. Meningitis comes in the dry season and cholera with the rains. The people most likely to die are small children with diarrhoea and old people alone under a metal roof. The ranked actions below are written for the region and its partners. These are the ones you can take yourself.

Keep oral rehydration salts (ORS) and zinc at home for every child under five by 30 November, and restock by 15 February

PARTIALLY CONFIRMED | FROM ACTION 1

BY WHEN	COST TO YOU	PROTECTS
30 November 2026; restock by 15 February 2027	low; the local price was not verified, so ask at the health centre (CSI) or a pharmacy	children under five

In a Sahelian population (Nouna, Burkina Faso), under-five deaths rose 3.7% for each 1 °C of heat. Diarrhoea is one of the main ways heat kills small children. ORS used early is estimated to prevent most diarrhoea deaths in under-fives; that estimate comes from a modelling review, not a trial in Maroua. Start ORS at the first watery stool, and go to the health centre if the child cannot drink, is very sleepy, or the diarrhoea is bloody.*

Treat and cover all drinking water, and give each person their own cup

PARTIALLY CONFIRMED | FROM ACTION 7

BY WHEN	COST TO YOU	PROTECTS
now, and every day through the hot season	a covered jerrycan and chlorine or bleach; local prices were not verified	the whole household, against cholera

* Suggested formulation: check with a health worker before sharing.

The 2010 cholera epidemic killed 602 people in the Far North, and a Far North study found that people shared cups at roadside water jars. In January 2026, households in Maroua were sharing unprotected water points with livestock while Camwater sent emergency tankers. Water that looks clean can still carry cholera.

In the hottest hours, wet your skin and clothes and rest in the shade; above about 38 °C, keep the fan off

CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
now through November, and again from March 2027	none	older people, outdoor workers, and anyone sick

In a 2024 laboratory trial at 45 °C and 15% humidity, close to Maroua's March and April, a fan increased strain on the heart, and wetting the skin with the fan off helped. Move errands and outdoor work out of the hours from 11:00 to 16:00 where you can. See "Do not do this" for the full explanation.*

Learn the signs of heatstroke and act fast: confusion, collapse or fits, and very hot skin

PARTIALLY CONFIRMED ROLE-SPECIFIC

BY WHEN	COST TO YOU	PROTECTS
now	none	everyone in the household

Move the person into shade, pour water over their skin, and take them to the nearest health centre or the Hôpital Régional de Maroua. Emergency numbers in Cameroon are 119 for medical help, 118 for the fire service, and 117 for the police. These numbers come from the Canadian government's travel advice, not from a Cameroonian source, so also keep the number of your health centre.*

Agree with one neighbour that you will check on each other at about 2 p.m. on very hot days

PARTIALLY CONFIRMED FROM ACTION 3

BY WHEN	COST TO YOU	PROTECTS
30 November 2026	none	

* Suggested formulation: check with a health worker before sharing.

		old people living alone, babies in metal-roofed rooms, and displaced families

The people who die in heat are usually indoors and alone, not on the street. In Rome, a programme that kept regular contact with isolated older people was linked with fewer than half the expected heat deaths. That is evidence from Italy, not from Cameroon, but the step costs nothing.

From December to June, take anyone with fever and a stiff neck to the health centre the same day

CONFIRMED FROM ACTION 2

BY WHEN	COST TO YOU	PROTECTS
December 2026 – June 2027	transport; ask about free treatment during an epidemic	children and young adults

The 1992 meningitis epidemic in northern Cameroon killed 968 people. The vaccine given since 2011 stopped serogroup A, but serogroup W still circulates, and early antibiotics are what save lives. A baby who is floppy or has a bulging soft spot also needs the health centre the same day.*

Keep a radio with spare batteries and a charged phone for power cuts, and listen to RFI on 101.6 FM for heat warnings

PARTIALLY CONFIRMED FROM ACTION 5

BY WHEN	COST TO YOU	PROTECTS
30 November 2026	batteries; local price not verified	the household's access to warnings during cuts

Load-shedding hit Maroua from 20 January 2026 because the Lagdo reservoir was low. RFI broadcasts in Maroua on 101.6 FM in French, Hausa, and Fulfulde. The national climate observatory, ONACC, issued heat warnings for the Far North in 2024 and 2026.

If you fast in Ramadan (expected around 8 February – 9 March) or Lent (from 10 February), drink plenty between sunset and dawn, and ask your imam or pastor about the exemptions for the sick, the elderly, and pregnant women

PARTIALLY CONFIRMED FROM ACTION 8

* Suggested formulation: check with a health worker before sharing.

BY WHEN	COST TO YOU	PROTECTS
February–March 2027	none	older people and anyone with a chronic illness

Ramadan falls inside the forecast March peak. In February 2026, ONACC warned of dehydration and heatstroke among people fasting in Maroua, with highs of 40–46 °C. In Karachi's 2015 heatwave, fasting itself was not found to be a significant risk factor, so treat this as sensible protection rather than a proven cause of death.

If you have cash to spare, buy your March–May sorghum by 15 December, while prices are low, and store it in airtight bags

PARTIALLY CONFIRMED · FROM ACTION 9

BY WHEN	COST TO YOU	PROTECTS
15 December 2026	hermetic bags about XAF 1,200–2,350 (US$2–4) each; no verified seller in the north, so ask agro-dealers	the household food supply in the lean season

At harvest, Maroua red sorghum was 38% below its five-year average, and FEWS NET projects it about 40% higher by May 2027. Only buy ahead if the cash is not needed for something more urgent, and keep stored grain safe.

What to ask of the people in charge

- Ask the head of your health district where the nearest rehydration point in your quarter will be, and whether it is stocked (action 1).
- Ask your chef de quartier whether the quarter's list of isolated people exists yet, and who checks on them (action 3).
- Ask the Governor's office whether the Hôpital Régional and the Camwater pumps are on a written list of lines protected from power cuts (action 6).
- Ask your commune when Camwater's scheduled tanker rounds start in your quarter (action 7).

* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. Put rehydration where people live: about 10 community ORS and zinc points across Maroua I, II, and III by 30 November, restocked by 15 February

PARTIALLY CONFIRMED | EMERGENCY MANAGER · PUBLIC HEALTH · HISTORIAN · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
DRSP Extrême-Nord and the Maroua health district chiefs	30 Nov 2026; restock by 15 Feb 2027	about XAF 3 million (US$5,100), an unsourced estimate; payer: the cholera response already funded by CERF	deaths from cholera and dehydration at home and on the way to care

Each point holds ORS, zinc, and Ringer's lactate, with a motorbike for referrals. A study of 97 fatal cholera-like cases in the Far North (2009–2011) found that 39% died in the community and 5% in transit. A national study of 2010 found that people who died lived farther from care and rarely used ORS at home. Cholera is active in the Far North in 2026: 585 cases and 19 deaths in the first three weeks of July. The points also serve the heat peaks, because dehydration is how heat kills small children.

* Suggested formulation: check with a health worker before sharing.

Transferability: the cholera evidence is from the Far North itself. The heat link rests on a study in Nouna, Burkina Faso (under-five deaths +3.7% per 1 °C), a similar Sahelian setting. The ORS effect is a modelled estimate. Cholera here peaks with the rains, so this is a cholera measure that also helps in heat, not mainly a March heat measure. CERF gave XAF 587 million (US$1 million) for the Far North cholera response in August 2026, so this should be paid from that, not from new money.

2. Prepare for meningitis before the season: stock ceftriaxone and rapid tests at every health post by 15 December, and have the emergency vaccine request ready

PARTIALLY CONFIRMED | HISTORIAN · ECONOMIST | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
DRSP Extrême-Nord, with WHO	15 Dec 2026, before the December–June season	not costed; outbreak vaccine is funded by Gavi through the ICG stockpile	meningitis deaths in children and young adults

The 1992 epidemic in northern Cameroon caused 8,046 cases and 968 deaths. The 2011 MenAfriVac campaign removed serogroup A, and no A has been isolated since 2014. Since 2013 the only serogroup isolated has been W, which that vaccine does not cover. A 2025 review says the meningitis A vaccine had not yet entered Cameroon's routine schedule, with Far North vaccination due to be completed in 2026. So the DRSP should check which children born since 2012 are protected, stock treatment, and prepare an International Coordinating Group request for the five-strain vaccine (Men5CV) if a district crosses the epidemic threshold.

* Suggested formulation: check with a health worker before sharing.

Transferability: this is the region's own record. The 2011 campaign's "101% coverage" was administrative and overstated, so actual protection among younger children is unknown.

3. Name the hidden households: by 30 November, Red Cross volunteers and each chef de quartier list the isolated and give each one a neighbour who checks at about 2 p.m. on hot days

PARTIALLY CONFIRMED | FRONTLINE · EMERGENCY MANAGER · PUBLIC HEALTH | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Cameroon Red Cross, Diamaré divisional committee, with chefs de quartier	30 Nov 2026; renewed by 15 Feb 2027	near zero (volunteer time)	deaths indoors among people nobody checks on

The list covers old people living alone, displaced people staying with host families, and infants and severely malnourished children in metal-roofed rooms. Each team includes at least one woman, because women care for the sick and some will not speak to male volunteers. The Far North holds 510,855 internally displaced people (IOM, July 2025). The Red Cross has a Diamaré delegation and warehouse in Maroua.

Transferability: the only outcome evidence is from Rome, where a programme of regular contact with isolated people over 75 was associated with heat mortality less than half the expected rate (IRR 0.44). That is an observational, high-income study. No tested low-income model exists, so treat this as transferred practice.

4. Get cash to the poorest households in February, before the March peak, through the Red Cross food-insecurity appeal

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST · FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Cameroon Red Cross, with IFRC	contracts in January; paid in February 2027	XAF 85.5 million (US$145,600) for 1,500 households, plus about 12% delivery	households cutting food, water, and care in the peak heat and the lean season

* Suggested formulation: check with a health worker before sharing.

IFRC Emergency Appeal MDRCM042 (Cameroon food insecurity) runs to 30 June 2027 and is 19% funded. In Yagoua in June 2026 it paid XAF 57,000 per household to 900 households. A top-up earmarked for Diamaré reaches 1,500 of the poorest and displaced households at the same rate. The list should be checked against the World Bank-financed safety-net project's (PFS-AIE) emergency transfer lists, so no household is paid twice.

Transferability: this works through payment systems the Red Cross has already used in the region in 2026. Cash beats food aid here because Maroua's markets work and grain is cheap. The Economist proposed releasing cash when Maroua-Salak station reaches 44 °C for three days. That is a proposal only: nothing like it exists in Cameroon, and there is no official record of how often Maroua reaches that temperature.

5. Pilot a heat alert that changes what the health system does: ONACC warning triggers a clinic heat protocol and cooling corners by 15 February

PARTIALLY CONFIRMED | PUBLIC HEALTH · HISTORIAN · EMERGENCY MANAGER · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
DRSP Extrême-Nord, using ONACC bulletins	**heat illness on the weekly surveillance return by 1 Dec 2026; cooling corners by 15 Feb 2027**	**about XAF 500,000 (US$850) per cooling corner, an estimate; pilot XAF 44 million (US$75,000), see funding**	**untreated heatstroke, and an invisible death toll**

On an ONACC heat warning, health posts switch to a heat protocol: triage for heat illness, extra checks on severely malnourished children, and messages on RFI's Fulfulde service. The Hôpital Régional and each district hospital keep a cooling corner with shade, water for wetting the skin, ice when power allows, and a written protocol. Heat illness and heat death are added to the weekly disease-surveillance (IDSR) return. ONACC already forecast 35–45 °C for the north in April 2024, and the Minister of Health issued heat advice that May, but no heat-health plan or warning system exists.

* Suggested formulation: check with a health worker before sharing.

Transferability: weak. The usual evidence, Ahmedabad's heat action plan, found an estimated 1,190 deaths averted a year, but its own authors say none of the differences were statistically significant, and Ahmedabad is a city of over 5 million with a municipal system. Treat this as a low-cost pilot and promise no mortality effect. Counting heat deaths matters on its own: without it, nobody will know whether any of these actions worked.

6. Keep the hospital and the water pumps powered: a written list of protected lines by 30 November, and diesel sized to the hospital's real use

PARTIALLY CONFIRMED | EMERGENCY MANAGER · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Governor of Extrême-Nord, with SOCADEL (formerly Eneo) and Camwater	30 Nov 2026	the list costs nothing; diesel for 30 days at 10 hours of cuts a day is roughly XAF 6–7.5 million (US$10,200–12,800), an estimate	the hospital and water supply failing during the peak

Load-shedding started on 20 January 2026 in Maroua and six other northern towns because the Lagdo reservoir was poorly filled. Backup comes from 72 MW of thermal plants limited by fuel money. In 2016–17, the press reported that "hospitals and prisons do not escape load-shedding." No priority-customer list for the north has been published. The utility was renationalised in May 2026 as SOCADEL, and its October tender for about 10 MW of emergency units has no commissioning date.

* Suggested formulation: check with a health worker before sharing.

Transferability: this is a local failure with a local fix. The diesel figure is an estimate. The first proposal of 3,000 litres would cover only 3–4 hours of cuts a day, so size the stock to the generator's measured use, at the regulated pump price of XAF 828 a litre.

7. Deliver water on a schedule from 1 February, and do not count on the cholera early-action protocol to pay for Maroua

PARTIALLY CONFIRMED | HISTORIAN · ECONOMIST · FRONTLINE · SKEPTIC | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Camwater, with the communes of Maroua I, II, and III	1 Feb 2027	not costed; unfunded	dry-season reliance on unsafe water, and cholera

In January 2026 Camwater sent an emergency tanker to Maroua, and households were sharing unprotected water points with livestock. The 3,000 boreholes promised in Maroua in 2011 were not built. Scheduled tanker rounds and water-point repairs should start before the March heat, not after a shortage. The Red Cross's cholera early-action protocol (approved March 2025, CHF 549,807) is triggered by floods or heavy rain, covers districts mainly in the Littoral, South-West, and Centre, and does not name Maroua. It cannot fire for a dry-season outbreak, so ask the Red Cross and the health ministry for contingency cover instead.

Transferability: this is local. The river-flood protocol (March 2026) covers Mayo-Danay and Logone-et-Chari only, not Diamaré.

8. Agree a Ramadan and Lent message with imams and pastors by 15 January

PARTIALLY CONFIRMED | FRONTLINE | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Imams of the Lamidat de Maroua and church leaders	15 Jan 2027	none	dehydration among people fasting in the March heat

Ramadan 2027 is expected from about 8 February to 9 March, depending on the moon, and Lent begins on 10 February. Both overlap the forecast March peak. In February 2026 ONACC warned of dehydration and heatstroke among people fasting in Maroua, Kousseri, Mora, and Yagoua. The message: drink plenty between sunset and dawn, know the signs of heatstroke, and remember the exemptions for the

* Suggested formulation: check with a health worker before sharing.

sick, elderly, and pregnant.

Transferability: in Karachi's 2015 heatwave, fasting was not a significant risk factor; low income and low education were. Frame the message as protective advice, not as a warning that fasting kills.

9. Buy sorghum while it is cheap: Office Céréalier tenders in November–December, released at cost from March to May

PARTIALLY CONFIRMED | FOOD SUPPLY CHAIN | ADAPTATION INVENTORY: NOT CHECKED

OWNER	BY WHEN	COST	PREVENTS
Office Céréalier, with MINCOMMERCE	buy by 31 Dec 2026; release March–May 2027	about XAF 103 million (US$175,400) for 1,000 tonnes; unfunded	a lean-season price spike that traders can exploit

At harvest, Maroua red sorghum was 13% below last year and 38% below its five-year average (FEWS NET, October 2026). FEWS NET projects it about 40% higher by May 2027, with the Far North in IPC Phase 3 from February. The wholesale price in May 2026 was about XAF 103,000 a tonne. MINCOMMERCE promotional sales of rice and oil before Ramadan would also help, through MIRAP if it operates in Maroua (not confirmed).

Transferability: the El Niño food signal here is weak. FEWS NET says El Niño is not expected to cause significant rainfall anomalies in Cameroon, and conflict, the naira, and aid cuts drive prices more. 1,000 tonnes is small against the region's need, and large NGO purchases have pushed up Far North prices before, so buy through tenders spread over time. No 2026 Office Céréalier operations were found.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	DRSP and health districts	Governor	Red Cross (Diamaré)	SOCADEL and Camwater	Communes and chefs de quartier	Imams and pastors	UN partners (WHO, UNICEF, WFP, IFRC)	Office Céréalier and MINCOMMERCE
1. Community ORS points	**A/R**	**I**	**R**		**C**	**I**	**C**	
2. Meningitis stock and vaccine request	**A/R**	**I**	**C**		**I**	**I**	**R**	
3. Lists of isolated people	**C**	**I**	**A/R**		**R**	**C**		
4. February cash		**C**	**A/R**		**C**		**R**	
5. Heat protocol pilot	**A/R**	**I**	**C**	**C**	**I**	**I**	**C**	
6. Protected power lines	**C**	**A**		**R**	**I**			
7. Scheduled water	**C**	**I**	**C**	**A/R**	**R**		**C**	
8. Ramadan and Lent message	**C**	**I**			**I**	**A/R**		
9. Sorghum purchase		**C**			**C**		**C**	**A/R**

What the chart shows: the DRSP is accountable for three of the nine actions and consulted on four more, so the health delegation carries this plan. The Red Cross is accountable for the two that reach households directly. Rows 3, 6, and 8 need a decision, not money.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

XAF 96.9 million (US$165,000)

TARGET

XAF 184.9 million (US$315,000)

CEILING

XAF 369.9 million (US$630,000)

LARGEST UNCERTAINTY

Nobody counts heat deaths here, so no placement's effect can be measured

Floor: the February cash alone, 1,500 households × XAF 57,000 = XAF 85.5 million (US$145,600), plus about 12% for delivery and targeting, rounded to XAF 96.9 million (US$165,000). It uses a channel that paid out in the region in June 2026. Target: the floor, plus a heat-protocol pilot of XAF 44 million (US$75,000: 30 facilities × US$1,500, plus US$20,000 for training and US$10,000 for alerts), plus XAF 44 million (US$75,000) for severe malnutrition treatment (500 children × about XAF 88,000, or US$150, all-in), giving XAF 184.9 million (US$315,000).

* Suggested formulation: check with a health worker before sharing.

Ceiling: cash for 3,000 households, XAF 193.8 million (US$330,000); the pilot at the panel's full scale, XAF 88.1 million (US$150,000); malnutrition treatment, XAF 44 million (US$75,000); and XAF 44 million (US$75,000) for water points only if the water and sanitation cluster confirms a gap beyond CERF's money. That gives XAF 369.9 million (US$630,000). The panel proposed a target of XAF 264.2 million (US$450,000); it fell by 30% because a water line duplicated committed CERF money and the per-facility cost was unsourced.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Cameroon Red Cross, via IFRC appeal MDRCM042	Action 4: cash to 1,500 households in Diamaré in February	XAF 96.9 million (US$165,000); up to XAF 193.8 million (US$330,000) for 3,000 households	top-up	XAF 57,000 per household, Yagoua, June 2026; appeal 19% funded	Confirmed
DRSP Extrême-Nord	Action 5: heat protocol pilot, cooling corners, and alerts	XAF 44 million (US$75,000); up to XAF 88.1 million (US$150,000)	direct grant	XAF 880,700 (US$1,500) per facility is an estimate; Maroua 2 has 11 health facilities and Maroua 3 has 10. No mortality effect is promised.	Partially confirmed
UNICEF supply pipeline, for severe malnutrition treatment	Action 5: the children at highest risk in heat	XAF 44 million (US$75,000)	prepositioned stock	Therapeutic food about XAF 27,400 (US$47) per carton (CHAI); about XAF 88,000 (US$150) per treated child all-in, an estimate	Partially confirmed
Water and sanitation partners, only if a gap is confirmed	Action 7: water points in hotspot quarters	XAF 44 million (US$75,000), ceiling only	direct grant	No published unit cost; CERF already funds the cholera response	Partially confirmed

The funder's one thing: no one funds heat here. The 2026 humanitarian plan was 26% funded in August, and the pre-arranged money in the Far North covers cholera and river floods only. November's deaths are largely beyond reach of new money; the March peak is where the cheapest lives can still be bought, if money is committed before January. Gavi already funds meningitis vaccine, so do not spend on it.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here.

Do not tell people to sit in front of a fan in the hot season, not even with wet skin *

CONTRAINDICATED CORRECTED

Above about 38 °C for people over 65 (37 °C for those on certain medicines, 39 °C for young adults), a fan blows hot air onto the body and adds heat (Morris, Jay, and colleagues, *Lancet Planetary Health* 2021). Maroua's March and April afternoons reach about 40 °C at 11–26% humidity, inside that zone. In a 2024 trial at 45 °C and 15% humidity, the fan increased strain on the heart, and a 2025 trial found that wetting the skin did not offset the fan's harm. The panel's own draft advice, and a proposal to fund solar fans, would have got this wrong.

Instead: wet the skin and clothes with the fan off, rest in shade, and drink. Use a fan only when the air is below about 38 °C, at night for example.

Do not just say "drink more water"

CONTRAINDICATED CONFIRMED

The water people can reach in the dry season is often shared with livestock or sold from open jars with shared cups. The 2010 cholera epidemic killed 602 people in the Far North, and cholera is active in the region again in 2026.

Instead: "drink treated, covered water from your own cup," with ORS at home and at the community points. *

Do not rely on text messages or posters for warnings

CONTRAINDICATED CORRECTED

Only about 17% of rural women in the Far North could read (ministry figure from 2012, the latest regional figure found). Women carry the water and care for the sick, so a written warning misses the people who act on it.

* Suggested formulation: check with a health worker before sharing.

Instead: radio (RFI's Fulfulde and French service on 101.6 FM), imams and pastors, chefs de quartier, community health workers, and short voice notes in Fulfulde.

Do not set up cooling centres that draw crowds to the central market at midday

CONTRAINDICATED CORRECTED

On 22 July 2015 at about 14:30, two suicide bombers struck the central market entrance and the Barmaré quarter, killing 13 people, including the bombers. No attack inside Maroua city was found in 2025–26, but attacks continue in Mayo-Sava. Check the current security advice before any public gathering is planned. A centre people must cross the city to reach in the midday heat also misses the people least able to travel.

Instead: shade and water at places people already go: health centres, mosques, churches, and moto-taxi stands.

WHAT HAS HAPPENED HERE BEFORE THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1992	Meningitis epidemic, northern Cameroon, January–May, during the 1991–92 El Niño	968 deaths, 8,046 cases (health ministry, unpublished, cited in 2013)	Mass MenAfriVac vaccination only in December 2011; serogroup A then disappeared
1998	Cholera, Extrême-Nord, June–October (rainy season)	107 deaths, 633 cases (case fatality 17%); 12 deaths in urban Maroua	Case fatality fell in later outbreaks as treatment was decentralised
2010	Cholera, Far North, from 6 May, the region's worst in over 20 years	602 deaths, 9,421 cases (88% of Cameroon's cases)	Temporary community treatment centres; water supply still not fixed
2012	Floods after the Maga dyke breach	37,816 displaced in the Far North; regional death toll not established	Flood early-action protocol in 2026 covers Mayo-Danay and Logone-et-Chari, not Diamaré

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2015	Twin suicide bombings, central market and Barmaré, 22 July	13 dead, including the bombers; 32 injured	Crowd security shapes any public gathering since
2024	March–April heat, the hottest since 1981 in the Far North (WFP), during the 2023–24 El Niño	No official count; reports of "many" deaths and of 50–60 burials a day are unsourced	ONACC warnings and ministry advice; no heat plan, and heat deaths still not counted
2024	Floods, August–October, worst in Mayo-Danay and Logone-et-Chari	30 deaths (OCHA, September); 365,060 affected	CERF XAF 2.35 billion (US$4 million) after the event
2026	Load-shedding from 20 January (low Lagdo reservoir); cholera from July	Cholera: 19 deaths by 22 July, 31 by 12 August, in the region	Eneo renationalised as SOCADEL; CERF XAF 587 million (US$1 million) for cholera

The recurrence nobody has noticed: the panel's historian proposed that Maroua's worst years come in the dry season right after an El Niño winter. Checked against NOAA's index, that does not hold. The 1998 and 2010 cholera epidemics came with the rains as La Niña set in, and 2010 was the deadliest year on this list. What El Niño years do bring here is dry-season heat and meningitis while the event is under way, as in 1992 and 2024. This winter and March are exactly that window. The water promise made in Maroua in 2011, 3,000 boreholes, has not been kept, and every dry season pays for it.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Fans help** versus **fans harm here**. The Philanthropist and Emergency Manager proposed fans, including solar fans, in cooling corners. The Skeptic and the verifier found that above about 38 °C in dry air fans add heat, even with wet skin. Measured indoor temperatures in Maroua homes and clinics during the hot season would settle when a fan helps.
- **Buy grain now** versus **give cash later**. The Food lens wants the Office Céréalier to buy 1,000 tonnes of cheap sorghum. The verifier found that markets work, cash is the better tool, and big purchases have pushed up prices before. Monthly FAO price data for Maroua would show whether a price spike is actually coming.
- **Release cash on a temperature trigger** versus **no record to set it on**. The Economist proposed paying when Maroua-Salak reaches 44 °C for three days. No official station record shows how often that happens, so the basis risk is unknown. A station history from the national

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meteorology service would settle it.

- **2,500 forecast deaths** versus **no local count**. Death registration in the Far North is low (one EU source, not independently checked, puts civil registration at 38%), and no heat death has been officially counted. The Public Health lens says the forecast cannot be tested here until heat deaths are recorded, which is why action 5 adds them to surveillance.

WHAT NOBODY COVERED

BLIND SPOT

No lens researched people held in institutions: the prison, Quranic boarding pupils, and patients in hospital wards without cooling. Each is a closed group in which heat and water shortages concentrate. No lens interviewed anyone in Maroua, and no quarter-by-quarter data on housing, water access, or displaced families was found, so the actions name the city's communes rather than particular quarters.

HOW TO READ THIS

METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are published at region level, here the whole Extrême-Nord Region of several million people. They are not city figures. The El Niño projection covers *heat only*; meningitis, cholera, flood, and food insecurity are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** This briefing was built on Angus Shaw's initiative, not from a request, as the first for Cameroon, the next country on the heat tracker's list without one. Extrême-Nord is Cameroon's worst region on the tracker (3,000 of 7,300 deaths), and Maroua is its capital. The location is the city of Maroua (communes of Maroua I, II, and III, Diamaré department), population about 300,000 in the city and about 527,000 in the urban area (UN, 2025). Mortality figures exist only at region level; city detail is used for hazard exposure and action design, not for death counts. Actions are not yet specific to individual quarters.

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- **Exposure.** Climate Impact Lab El Niño Heat Mortality Tracker data, file `v261026_adm1_output_BASE_rounded_cust.csv` (version of 26 October 2026, pre-peer-review): Extrême-Nord Region (GADM CMR.4_1), 490 additional heat deaths June–September 2026, 2,500 forecast October 2026 – March 2027 (October 590, November 510, December 120, January 140, February 200, March 910), 3,000 in total; Cameroon, 7,300; the world, 857,400. This update supersedes the September report *451,000 at Risk*. Method: ECMWF SEAS5 seasonal forecasts applied to temperature–mortality relationships against a 1996–2025 baseline. The figures count heat only; we inferred this from the data, because cold countries show exactly zero in every winter month. They are never added to the Impact Map's long-run projections.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions were also checked for availability in Cameroon, a named payer, a trigger, and whether they already exist. Food actions were checked for published price series, the timing of the price peak, delivery capacity, and cash versus in-kind. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. A separate verifier checked every event in the record against its source and NOAA's El Niño index.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Could not be consulted: the inventory loads its entries in the browser and returned none to the research agents. No action is tagged with a precedent from it.
- **Money.** Costs are shown in Central African CFA francs (XAF) with US dollars in parentheses, at XAF 587.14 = US$1. The rate comes from the fixed peg of XAF 655.957 = €1 and the ECB reference rate of €1 = US$1.1172 (30 October 2026). Figures published in Swiss francs (Red Cross protocols) are left in francs.

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- **Reader.** No role was given, so the For you section is written for a resident. Its steps were taken from the verified actions and from what the panel found about households here; steps not drawn from a ranked action were verified on the same three tests. The rest of the report is written for the region and its partners.
- **Health wording.** Passages where a wording error could hurt someone (ORS and zinc, cooling and fans, heat-illness first aid and the 119 emergency number, meningitis danger signs, and safe drinking water) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number. Estimates are labelled as estimates. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in Maroua was interviewed. The 2024 heat temperatures (34.9 °C regional mean, 45–46 °C in the shade) could not be opened at source. No official heat death count exists. Demographic and Health Survey files returned errors, so the female literacy figure is from 2012. The current security situation inside Maroua was not confirmed from an official source.

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Cartwright et al., Epidemiol Infect 2013 · confirmed
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WHO, cholera in Cameroon, 15 Oct 1998 · confirmed
UNICEF Cameroon, cholera flash update no. 2, July 2026 · confirmed
Munos, Fischer Walker, Black, ORS and diarrhoea mortality, IJE 2010 · confirmed (modelled)
Diboulo et al., heat and mortality in Nouna, Global Health Action 2012 · confirmed
Meningitis 1992 and MenAfriVac, Pathog Glob Health 2013 · confirmed; serogroups after 2013, PMC8490062 · confirmed; routine introduction, PAMJ 2025 · confirmed; Gavi meningococcal support · confirmed
Hess et al., Ahmedabad heat action plan, 2018 · confirmed (not significant)
ONACC heat forecast, Cameroon Tribune, April 2024 · confirmed; ONACC fasting warning, Journal du Cameroun, Feb 2026 · confirmed; Minister of Health heat advice, May 2024 · confirmed
Rome isolation programme and heat mortality, PMC5923757 · confirmed (observational)
OCHA, humanitarian dashboard, 31 Dec 2025 · confirmed; Red Cross in Maroua, Cameroon Tribune · confirmed
IFRC, appeal MDRCM042 · confirmed; Yagoua cash, Cameroon Tribune, June 2026 · confirmed; World Bank PFS-AIE additional financing · confirmed

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IFRC cholera early-action protocol, Anticipation Hub · confirmed; river-flood protocol, MDRCM044 · confirmed
CERF, 2024 flood allocation · confirmed; cholera allocation 2026, report · confirmed; humanitarian plan funding, Cameroon Tribune, Aug 2026 · confirmed
Load-shedding 2026, Business in Cameroon · confirmed; renationalisation, Africanews · confirmed; 2016–17 cuts, camer.be · confirmed; diesel price, Cameroun24 · confirmed
Water and the 2011 borehole promise, AllAfrica, Jan 2026 · confirmed
Fans in heat: Morris et al., Lancet Planet Health 2021 · confirmed; NEJM 2024 · confirmed
Ramadan 2027 dates, Gulf News · confirmed (projection); fasting in Karachi 2015, Prehosp Disaster Med · confirmed
FEWS NET Cameroon, October 2026 outlook and June 2026 outlook · confirmed; sorghum price, FAO FPMA via aggregator · confirmed; Office Céréalier, Ecomatin · confirmed; NGO purchases and prices, Business in Cameroon · confirmed; WFP funding gap, WFP · confirmed; hermetic bags, TechXLab · confirmed; RUTF price, CHAI · confirmed
2015 bombings, Presidency of Cameroon · confirmed; 2012 floods, ReliefWeb · confirmed; 2024 floods, Journal du Cameroun citing OCHA · confirmed; 2024 heat, WFP factsheet · source not opened (403)
Literacy, MINPROFF 2012 via Coordination Sud · confirmed; RFI in Maroua, RFI 2026 · confirmed; emergency numbers, Government of Canada · confirmed (secondary); heatstroke signs, WHO · consulted
Exposure data · Climate Impact Lab, 451,000 at Risk (superseded) · Impact Map · Adaptation Inventory
Generated 30 October 2026 for Maroua, Extrême-Nord. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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